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Palaeomagnetism of the Ferrar dolerite in the northern Prince Albert Mountains (Victoria Land, Antarctica)

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SUMMARY

During the sixth Italian expedition to Victoria Land (Antarctica), Jurassic Ferrar dolerite sills were sampled at 23 sites in the northern Prince Albert Mountains. The rock magnetic properties were similar in all sites. Saturation remanence, remanent coercive force and Curie-temperature values pointed to titanomagnetite as the main carrier of magnetization. Thermal and AF demagnetizations were both effective in removing weak, secondary magnetization components below 250 °C and 20 mT peak field. Further demagnetization resulted in single-component, stable ChRM. Palaeomagnetic directions were derived after AF demagnetization at 20 mT, and site-mean VGPs computed for all sites but one (Thern A), where systematic changes of the ChRM inclination have resulted in single-sample VGPs dispersed astride the equator. The site-mean VGPs distribution is clearly bimodal: 15 VGPs cluster around a palaeopole (lat. 48°S, long. 226°E) in good agreement with the literature data, whereas seven VGPs are displaced toward a palaeopole at higher latitude (lat. 79°S, long. 181°E).

All VGPs can be regarded as Jurassic, because no evidence of substantial secondary components was found, and experimental data showed that the PTVRM (partial thermal viscous remanent magnetization) acquisition rate is too low to account for complete remagnetization of the rocks giving the anomalous VGPs. The Thern A and the southward displaced VGPs plot along a great circle through the East Antarctica Jurassic palaeopole. As a high reversal rate in the Jurassic is well documented by magnetostratigraphy, and the reversal VGP path is often confined to a narrow longitudinal band, we suggest that the great circle is a palaeomeridian and the sills characterized by anomalous VGPs were intruded during transitional phases of the geomagnetic field. Similar anomalous VGPs previously reported from the Ferrar Supergroup rocks of Northern Victoria Land also plot along the same palaeomeridian, which is only 30° away from the palaeomeridian derived from the Jurassic intrusives of the Dufek Massif in the Transantarctic Mountains (Burmester & Sheriff 1980).

The Jurassic palaeopole of East Antarctica has been recalculated from 103 VGPs available in the literature. The position is 57°S, 221°E when all VGPs are taken into account, and 51°S, 223°E when the high-latitude VGPs from Victoria Land are discarded as anomalous.

Key words: Antarctica, Ferrar dolerite, Jurassic, palaeomagnetism, palaeopole.

INTRODUCTION

A huge volume of Jurassic continental flood basalts characterizes the early phases of the break up of Gondwana. Basalt flows and dolerite sills have been recognized in southern Africa, Antarctica and southern Australia, and have been extensively studied as palaeomagnetic markers of the drifting of these continents. The Jurassic igneous rocks of Antarctica are known as the Ferrar Supergroup. This Supergroup extends for some 4000 km along the Transantarctic Mountains and encompasses the Kirkpatrick basalts, the Ferrar dolerite sills and the gabbroic Dufek intrusion. Palaeomagnetic studies on these rocks began in the fifties (see references in Table 2), and have yielded consistent

results. The virtual geomagnetic poles (VGPs) from the pre-1984 literature cluster around a palaeopole located at lat. 53°S and long. 222°E, which is in good agreement with Jurassic palaeopoles from the other land masses of Gondwana. However, subsequent work on the Ferrar Supergroup rocks in northern Victoria Land (Delisle & Fromm 1984, 1989; McIntosh, Kyle & Sutter 1986) has disclosed VGPs at higher latitudes (64°S–78°S). A comprehensive explanation of this discrepancy has yet to be found.

During the sixth Italian Antarctic expedition to Victoria Land (austral summer 1990/91), Ferrar dolerite sills were sampled along a 400 km sector of the northern Prince Albert Mountains, running northwards from the David Glacier to the Campbell Glacier. The 23 visited sites (Fig. 1) have provided palaeomagnetic data that fill the gap between the

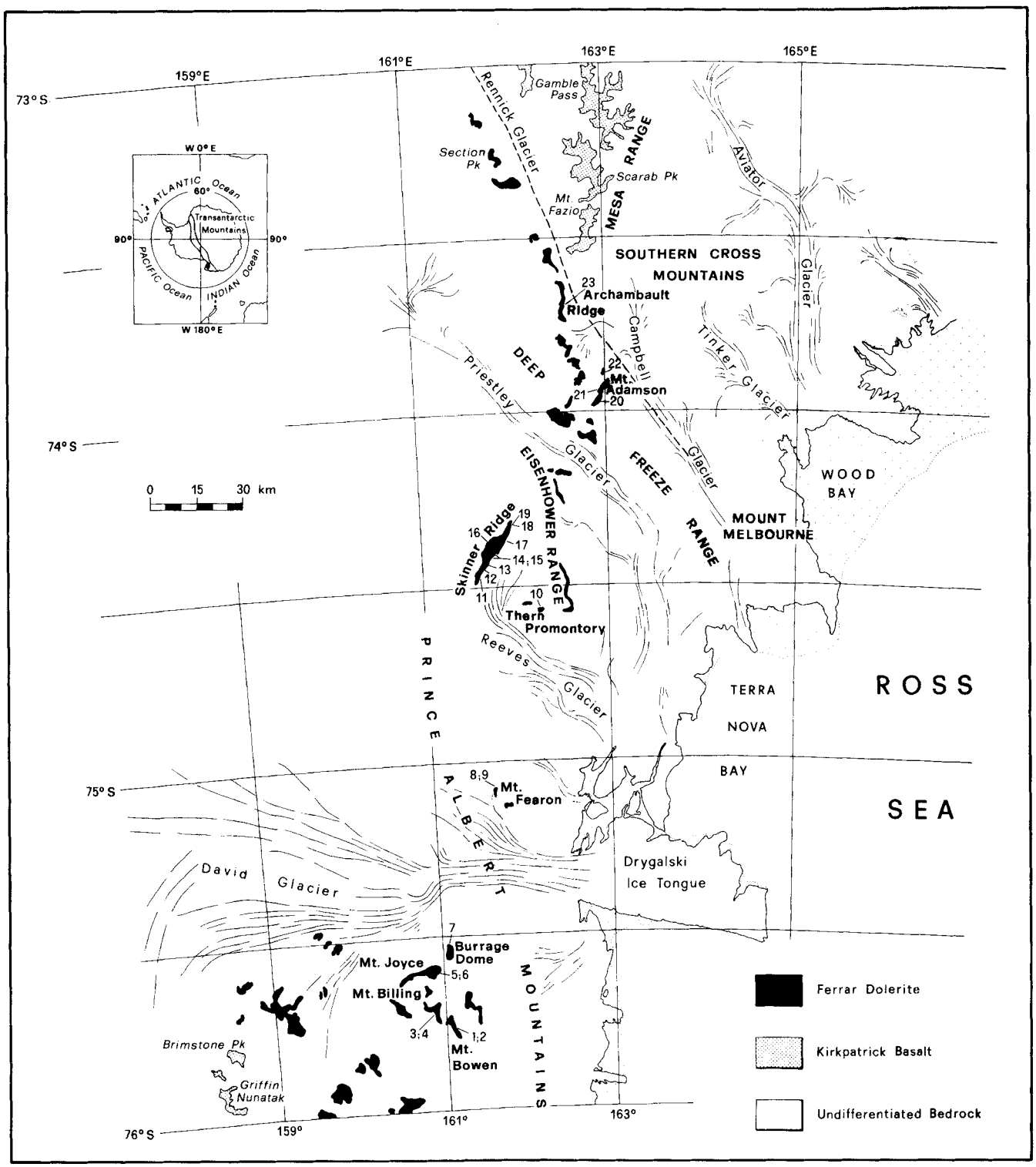


Figure 1. Geological sketch map of the northern Prince Albert Mountains region and location of the sampling sites (1–23).

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localities investigated by Cherry & Noltimier (1981) south of the David Glacier (Brimstone Peak and Griffin Nunatak) and those studied by McIntosh *et al.* (1986) in the Rennick Graben (Mesa Range).

GEOLOGICAL SETTING AND SAMPLING

The region is part of the Wilson Terrane of Northern Victoria Land, where two main rock sequences crop out, a pre-Cambrian–Lower Palaeozoic basement and an Upper Palaeozoic–Mesozoic cover, separated by the Kukri unconformity. The basement consists of high-grade metamorphics, intruded by Ordovician granitoids associated with the Ross orogeny. The cover rests unconformably on the Kukri peneplain and consists of Permo-Triassic continental sediments (Beacon Group) and Middle Jurassic tholeiites (Ferrar Supergroup), subdivided into the Kirkpatrick basalts and the Ferrar dolerites, which intrude both the basement and Beacon Group. No major tectonic event occurred in Victoria Land after Jurassic magmatism until the uplift of the Transantarctic Mountains, which began some 45 million years ago, and the Cainozoic to Recent alkaline magmatism along the Ross Sea margin.

The tectonic setting, petrology and geochronology of the Ferrar Supergroup have been widely investigated (Kyle, Elliott & Sutter 1981; Elliot, Siders & Haban 1986; Elliot & Foland 1986; Roland & Tessensohn 1987; Elliot 1991; Faure & Mensing 1993). On a regional scale, the Kukri peneplain dips 2°–5° westwards and the only local increase is observed adjacent to the faults along the shoulders of the Rennick Graben. The Kirkpatrick basalts must at least have been as thick as the 800 m existing in the Mesa Range. The thickness of individual Ferrar dolerite sills varies between a few metres and 400 m. K-Ar ages of the magmatism range from 180 to 90 Ma. The best age estimate for the Kirkpatrick basalt is 180 ± 5 Ma (Elliot & Foland 1986), the younger ages being explained as the result of argon loss. $^{40}Ar/^{39}Ar$ ages of 174 ± 1 Ma (McIntosh *et al.* 1986) and 177 ± 2 Ma (Fleming *et al.* 1991) have been obtained in the Mesa Range. The Ferrar dolerite is believed to be somewhat younger than the Kirkpatrick basalts, because evidence has been found of sills intruding the basalts and these, in their turn, must have contributed to the overburden needed for sill emplacement. A younger episode of sill intrusion may have occurred at ≈165 Ma (Elliot & Foland 1986).

The geological maps of the GANOVEX team (1987) and Carmignani *et al.* (1988) indicate that our sampling sites (Fig. 1) are all Ferrar dolerite intrusives. Skinner & Ricker (1968), however, consider the outcrops at the top of Mount Joyce (our site 6) as extrusive Kirkpatrick basalt. Even if no detailed petrographical analyses were done, at least one thin section for each site was examined. The rocks are medium to fine grained and have a subophitic texture consisting of plagioclase, clinopyroxene, subordinate orthopyroxene, opaques and occasionally olivine.

Four or five hand samples were collected from each site, and oriented by both magnetic and, with the exception of site 23, sun compass. The local values of the magnetic declination vary from ≈130°E in the northern sites to ≈150°E in the southern sites. They are in good agreement with the regional trend of the geomagnetic field, with the sole exception of the value (92°E) measured at site 13. A total of 138 hand samples were collected. Three cores were cut from each sample and one or two standard specimens from each core. Three specimens (one from each core) were measured, and each specimen was given unit weight to compute the site mean values of the magnetic parameters.

MAGNETIC MEASUREMENTS

Magnetic mineralogy was investigated by isothermal remanent magnetization (IRM) and susceptibility versus temperature experiments, using a Bussi pulse magnet and a Bartington MS2/WF equipment respectively. The IRM curves were all alike (Fig. 2): remanence increased steadily with the applied field and saturation was reached within field values of 0.10–0.15 T. The remanent coercive force was in the range 20–30 mT. The susceptibility versus temperature curves initially displayed a gradual increase. A steep decrease occurred above 500 °C, and the Curie temperature values ranged between 540 °C and 565 °C. These results point to low-Ti titanomagnetite as the main magnetic carrier in the Ferrar dolerite.

Magnetic susceptibility k_m and natural remanent magnetization (NRM) of all specimens were measured first. The k_m value ranged between 1203×10^{-6} and 53305×10^{-6} SI, and the NRM intensity between 0.11 and 6.49 A m^{-1} (Table

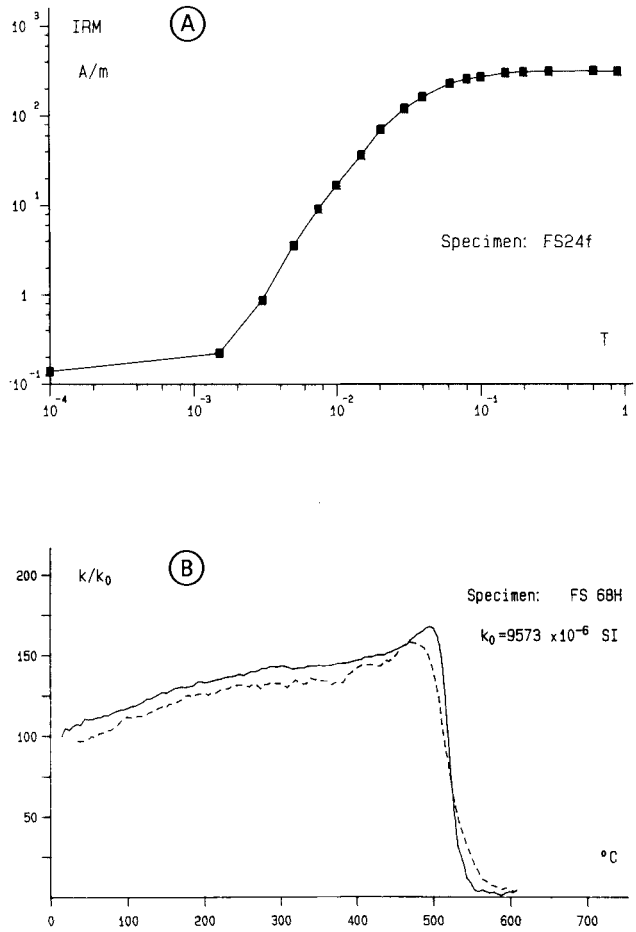


Figure 2. (a) Isothermal remanent magnetization (IRM) acquisition curve. (b) Normalized susceptibility versus temperature curve. Symbols: full/dotted line = heating/cooling cycle; k_0 = bulk susceptibility at room temperature.

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1). The remanent magnetization is higher than the induced, which was calculated from the k_m values and a mean regional geomagnetic field of 53 A m^{-1}. The Koenigsberger ratio varied between 1 and 5.5; remanence must thus be taken into account as the main source of geomagnetic anomalies produced by the Ferrar dolerite (Bosum *et al.* 1988). The NRM directions were always of normal polarity and tightly grouped. The mean direction was not computed for two sites: Thern A, because of systematic inclination changes (see next chapter), and Skinner Ridge 2. The NRM directions from this site were randomly scattered, and a magnetic anisotropy study showed that even the principal susceptibilities directions were scattered; very bad weather during the sampling probably resulted in severe orientation errors.

At least one specimen from each site was stepwise demagnetized in peak alternating field up to 100 mT, and one thermally up to 600 °C. No heat-induced mineralogical transformation occurred, because there was no significant change in the magnetic susceptibility measurements after each temperature step. Small directional changes only occurred in the first demagnetization steps, below 20 mT or 250 °C. The Zijderveld plots (Figs 3 and 4) show that very weak secondary components were removed at first, then the characteristic remanent magnetization (ChRM) was isolated and directions went toward the origin. The normalized intensity decay curves for thermal demagnetization (Fig. 3) showed a sharp decrease within a range of some tens of degrees (350–400 °C) and a gradual one up to the Curie point. As directions did not change throughout the blocking temperature spectrum, a primary origin of the ChRM can be inferred. The decay curves for AF demagnetization (Fig. 3) point to median destructive fields in the range 20–40 mT.

AF demagnetization at 20 mT peak value was chosen as a routine procedure for all specimens, and slightly reduced the within-site scatter at most sites (ChRM columns in Table 1). The ChRM directions were used to compute the VGP positions. No tectonic correction was made, because this would only have been possible at the few sites where direct evidence of the Kukri peneplain was available. Moreover, the gentle dip of the peneplain may well be a primary topographic feature and not related to tectonic events. The

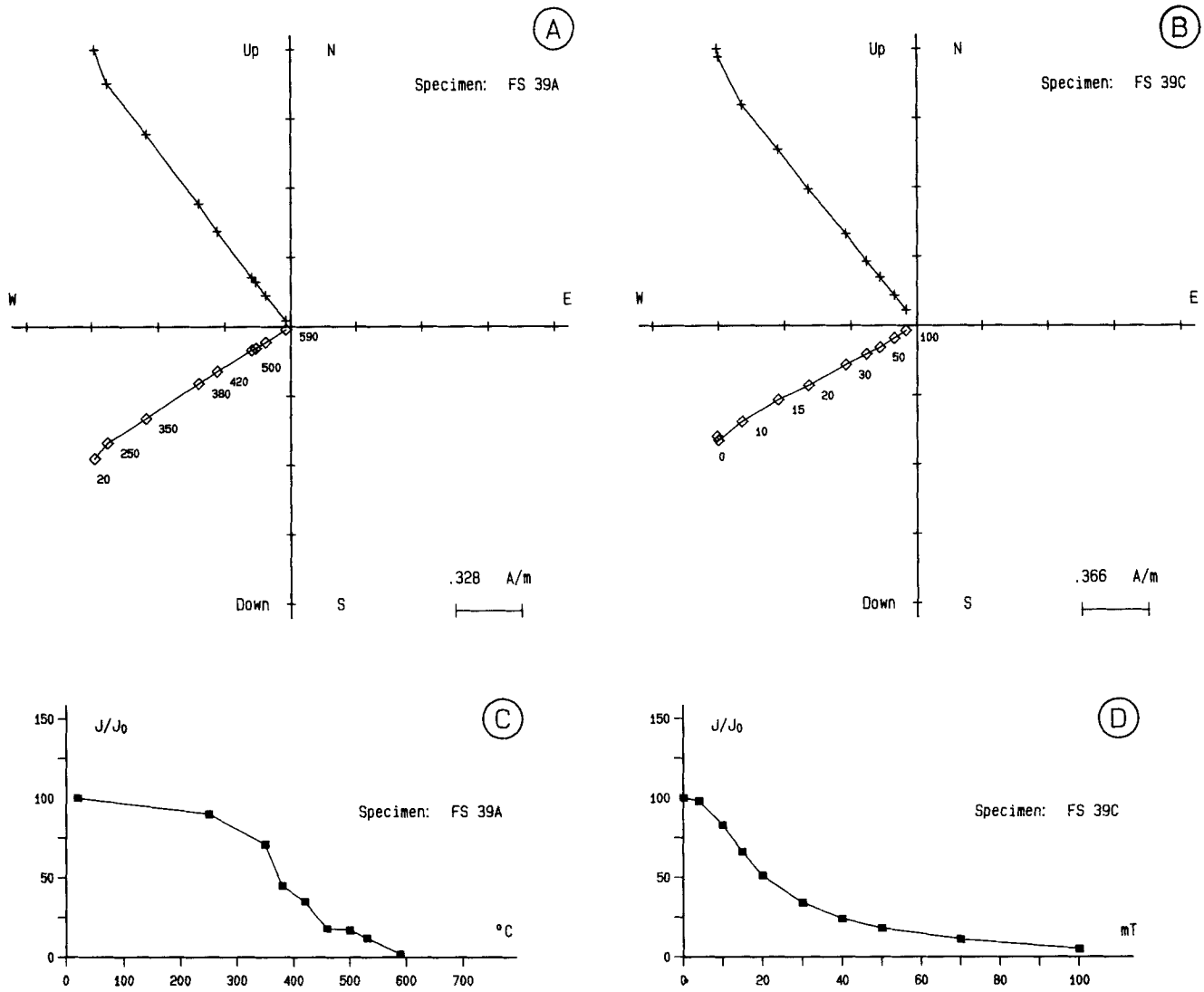


Figure 3. Zijderveld diagrams (A, B) and normalized-intensity-decay curves (C, D) for thermal and AF demagnetization (specimens from site 8). Symbols: open squares = declination; crosses = apparent inclination; figures = temperature value in °C, peak-field value in mT.

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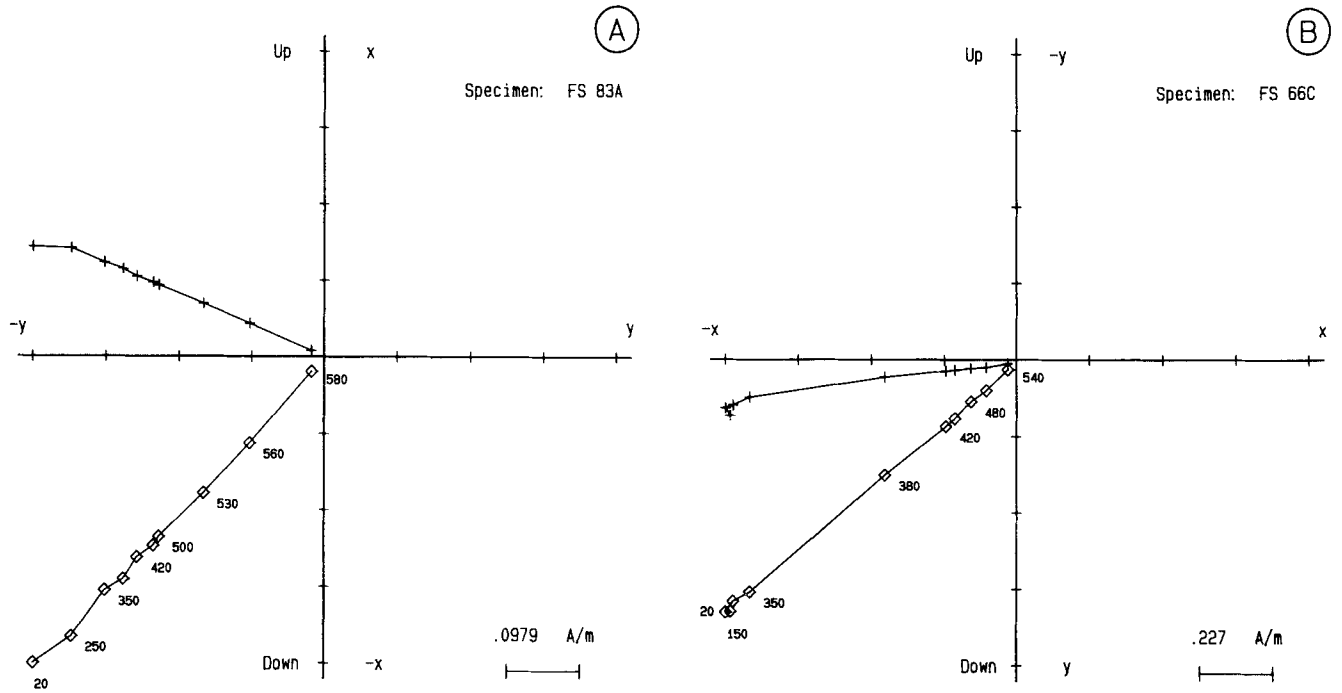


Figure 4. Zijderveld diagrams for thermal demagnetization (specimen A from site 1, specimen B from site 10, Thern A). Projection in the reference system of the specimen, other symbols as in Fig. 3.

VGPs are plotted as southern poles in Fig. 5. A bimodal distribution is evident: 15 VGPs cluster around lat. 48°S, long. 226°E ($K = 49$, $A_{95} = 5.5°$), seven around lat. 79°S, long. 181°E ($K = 106$, $A_{95} = 5.9°$).

The first position is in good agreement with the palaeopole derived from the pre-1984 literature data (lat. 53°S, long. 222°E). The second is displaced towards the pole, not far from other anomalous VGPs from Northern Victoria Land (Delisle & Fromm 1984, 1989; McIntosh *et al.* 1986). The assumption that the single component, stable ChRM is the primary one therefore needs further examination. Complete remagnetization could have been the result of

Table 1. Palaeomagnetic results from Ferrar dolerite, northern Prince Albert Mountains (Victoria Land, Antarctica).

	Site		Coordinates		n	km (SI x10⁶)	Jr (A/m)	NRM			ChRM				VGP	
			Lat. S	Long. E				D	I	α_{95}	D	I	α_{95}	k	Lat. S	Long. E
1	Mt. Bowen 1		75.75	161.05	15	4729	1.34	235.5	-87.7	1.4	227.2	-87.2	1.7	524	72	174
2	Mt. Bowen 2		75.75	161.03	15	1203	0.11	268.1	-87.6	2.7	273.4	-87.5	2.5	242	75	181
3	Mt. Billing 1		75.71	160.91	15	6814	0.55	335.1	-86.1	2.8	345.9	-87.3	2.4	264	81	169
4	Mt. Billing 2		75.71	160.88	15	10560	1.83	284.6	-86.2	3.3	303.8	-84.7	2.3	285	78	207
5	Mt. Joyce 1		75.62	160.84	15	13001	3.77	259.0	-72.4	5.9	257.9	-71.1	6.0	41	50	221
6	Mt. Joyce 2		75.61	160.80	15	13321	1.49	259.5	-83.1	8.9	260.0	-75.3	3.1	151	57	218
7	Burrage Dome		75.56	161.07	12	9124	1.42	285.2	-88.4	4.5	268.6	-86.5	4.3	105	74	187
8	Mt. Fearon 1		75.10	161.68	12	10513	1.51	265.1	-66.4	6.5	265.9	-63.6	6.4	48	42	233
9	Mt. Fearon 2		75.09	161.70	12	7576	0.69	254.3	-72.6	3.7	264.6	-68.3	3.1	193	48	229
		A			63	10237	1.55	---	---	--	---	---	--	---	--	---
10	Thern Promontory		74.58	162.25												
		B			53	8890	0.55	253.2	-74.5	6.8	243.3	-66.4	4.4	22	40	213
11	Skinner Ridge 1		74.49	161.57	15	16451	3.04	318.1	-88.2	1.5	313.4	-88.9	2.0	361	76	168
12	Skinner Ridge 2		74.45	161.64	9	32858	5.93	---	---	--	---	---	--	---	--	---
13	Skinner Ridge 3		74.44	161.67	11	53305	6.49	27.4	-74.6	3.8	3.3	-80.2	2.7	286	86	359
14	Skinner Ridge 4		74.42	161.73	15	9816	0.89	259.3	-75.9	2.7	247.0	-71.8	1.5	620	48	211
15	Skinner Ridge 5		74.41	161.75	18	20749	0.93	258.1	-74.6	3.9	261.7	-66.5	2.8	153	45	227
16	Skinner Ridge 6		74.39	161.75	15	7428	3.79	277.7	-60.4	7.1	275.0	-61.4	5.6	48	42	242
17	Skinner Ridge 7		74.37	161.88	12	6537	1.00	256.3	-72.1	2.7	256.9	-72.0	2.4	320	51	219
18	Skinner Ridge 8		74.34	161.92	12	10807	2.07	278.0	-73.7	5.5	276.0	-71.9	5.3	68	55	234
19	Skinner Ridge 9		74.33	161.92	12	33332	2.47	254.0	-72.6	3.9	245.5	-69.7	3.7	137	45	212
20	Mt. Adamson 1		73.98	162.95	15	19986	1.19	269.4	-68.8	3.1	244.2	-61.9	4.2	84	35	216
21	Mt. Adamson 2		73.94	162.95	12	14112	1.02	296.0	-68.4	3.8	296.3	-67.4	2.6	276	55	259
22	Mt. Adamson 3		73.88	162.95	15	15801	2.07	242.9	-72.1	3.1	241.9	-66.3	2.0	365	39	212
23	Archambault Ridge		73.69	162.59	9	11602	0.91	303.3	-67.5	9.3	285.5	-67.4	8.0	42	52	248

Abbreviations: n = number of specimens; k_m bulk susceptibility; J_r = NRM intensity; NRM = natural remanent magnetization; D, I = declination, inclination; α_{95} = semi-angle of confidence; ChRM = characteristic remanent magnetization after AF treatment (20 mT); k = Fisher's precision parameter.

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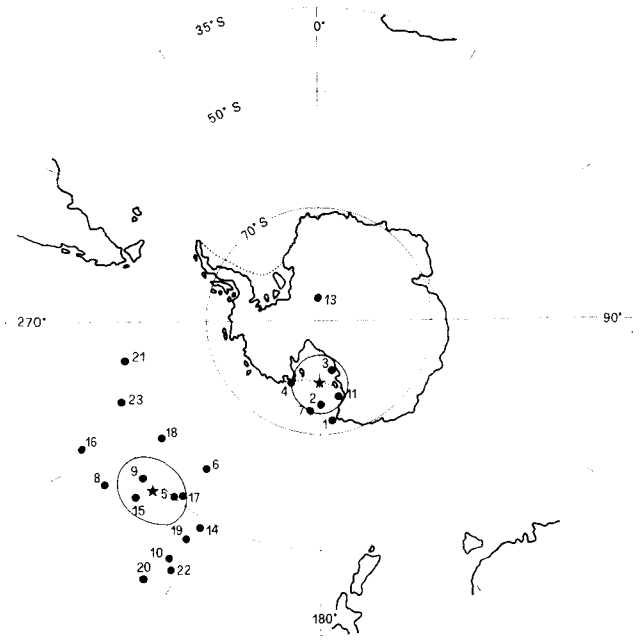


Figure 5. VGP positions from Ferrar dolerite sills in the northern Prince Albert Mountains. Figures = site number as in Table 1; stars = palaeopole with A_{95} ellipse of confidence.

long exposure to both a magnetic field and high temperature. The propensity of the Ferrar dolerite to acquire such PTVRM (partial-thermal-viscous remanent magnetization) was investigated by an experiment according to the technique described by Pullaiah *et al.* (1975).

Five samples from sites giving anomalous VGP were studied. Two specimens were chosen from each sample, in such a way that the NRM was roughly parallel to the specimen y-axis. The specimens were first AF demagnetized at 10 mT peak field. The twin specimens were then placed in a non-magnetic furnace, with the y-axis in opposite directions with respect to the magnetic field in the laboratory (0.045 mT), and heated at 250 °C at four time

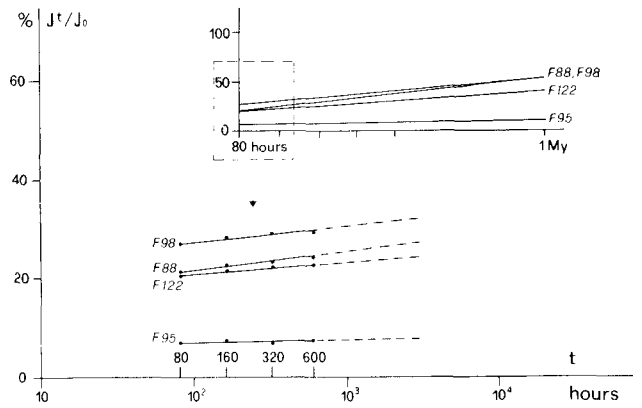


Figure 6. Normalized PTVRM intensity curves. J_0 = NRM intensity after 10 mT peak-field demagnetization; J_t = intensity after heating over time intervals of 80, 160, 320, 600 hours (temperature = 250 °C, applied field = 0.045 mT). The curves in the upper part show linear extrapolation of the experimental results to 1 Myr. Specimens F88, F95, F98 and F122 were collected at sites 2, 3, 4 and 12 respectively. The curve for specimen F86 (site 1), not shown in the figure, coincides with the specimen F98 curve.

steps (80, 160, 320, 600 hr). After each step, remanence was measured and the PTVRM calculated from each couple of specimens. Susceptibility measurements showed that no mineralogical changes occurred.

The results (Fig. 6) show a PTVRM increase proportional to log time, as expected from theory. Linear extrapolation to 1 Myr, i.e. to an order of magnitude comparable to the duration of geological processes, gives a PTVRM less than the initial remanence of the samples. The results of such extrapolation can be only considered as a first approximation (Pullaiah *et al.* 1975; Dunlop 1983). However, as the extrapolated PTVRM values range from 7 per cent to 55 per cent of the initial remanence, we may reasonably conclude that there is no experimental support for the hypothesis that the sills characterized by anomalous VGPs were completely remagnetized.

THE THERN PROMONTORY SECTION

The Ferrar dolerite sills are up to some hundreds of metres thick and must have taken some hundreds of years to cool down. Since the different parts of a sill would have passed the Curie point threshold at different times, palaeosecular variation and geomagnetic excursions could have been recorded. This hypothesis was tested by detailed sampling of the Thern Promontory sill, both because it is well exposed and because its petrology was investigated by Brotzu *et al.* (1988). During this expedition, snow thickly covered the previously studied section and sampling was carried out on a ridge 5 km to the east of Thern Promontory. The two sections are readily correlated, because in both of them dolerite overlies Beacon sandstones resting on the Kukri peneplain. A total of 39 hand samples were taken over a thickness of about 300 m and three specimens were measured for each sample.

Pilot specimens were used for IRM measurements, susceptibility versus temperature experiments, and AF and thermal demagnetization. The results of all tests were fully comparable to those obtained from other sites. The ChRM directions after AF demagnetization at 20 mT peak field are shown in Fig. 7. Declination shows some random variations throughout the section, while inclination is systematically different. Inclination is stable (≈74°) and similar to that from other sites in the lower part of the sill (Thern B), while in the upper part (Thern A) it shows a wide excursion between −53° and +37°.

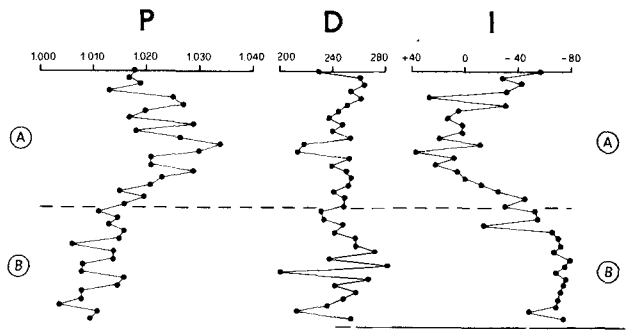


Figure 7. Anisotropy degree (P) and ChRM declination and inclination (D, I) curves from the Thern Promontory section. Samples are arranged in stratigraphical order and the dotted line separates the lower (B) and upper (A) units.

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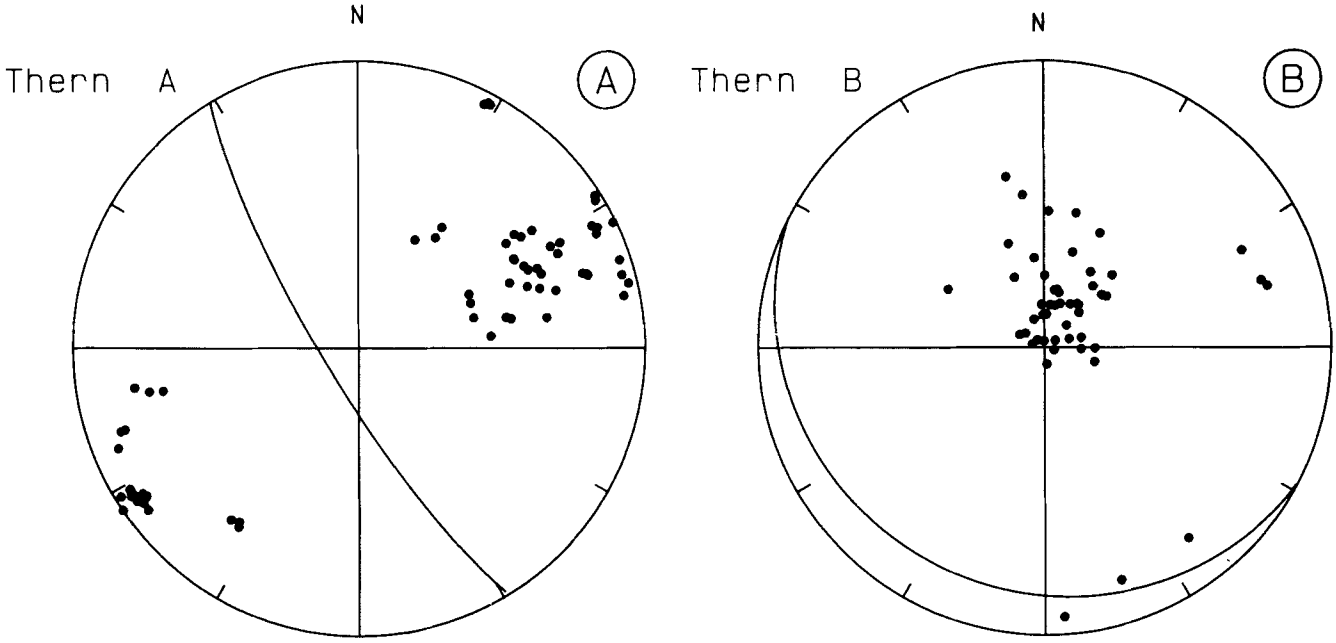


Figure 8. Equal-area projection of the magnetic foliation poles from the two units at Thern Promontory (great circle = site-mean magnetic foliation plane).

Demagnetization tests show that both in the lower and upper part of the section there are no significant secondary components and the ChRM direction is stable up to the Curie point (Fig. 4b). Remanence may therefore be considered as primary, and its direction a record of geomagnetic field during the cooling.

The Thern Promontory section was further investigated by measuring the anisotropy of magnetic susceptibility (AMS). All specimens were measured with a KLY-2 bridge, and the principal susceptibilities ($k_{max} > k_{int} > k_{min}$) and the corresponding directions of the ellipsoid axes were determined. The results (Figs 7 and 8) show that two different magmatic units crop out along the section. The anisotropy degree $P = k_{max}/k_{min}$, which indicates the degree of preferred orientation of the magnetic grains, is higher in Thern A, and lower in Thern B. The magnetic foliation, i.e. the plane defined by the k_{max} and k_{int} directions, is close to the horizontal in Thern B, steeply inclined in Thern A.

The different magnetic fabric can not be regarded as the source of the systematic changes in the ChRM inclination, because no correlation was found between the remanence direction and the magnetic foliation plane. Moreover, the P values were too low ($P \leq 1.034$) to have deflected the remanence vector from the geomagnetic field direction (Hrouda 1982).

Two different flow units therefore crop out at the Thern Promontory section, and the upper one recorded a varying geomagnetic field. Most southern VGPs calculated for each sample from Thern A are confined to a narrow longitudinal band (220°E–240°E), over latitudes from 20°S to 20°N (Fig. 9). As longitudinal patterns have been observed in the VGP path during polarity transitions (Clement & Kent 1985; Herrero-Bervera *et al.* 1989; Tric *et al.* 1991b), we suggest that during the slow cooling of the magma the Thern A sill recorded part of a Jurassic geomagnetic excursion or polarity transition.

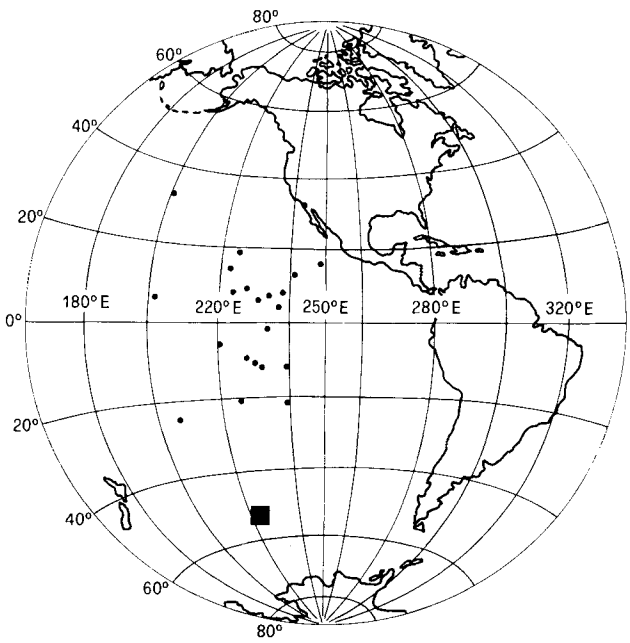


Figure 9. VGP positions from the Thern A sill. Dots = single-sample VGPs; square = Jurassic palaeopole of Antarctica from pre-1984 literature data.

DISCUSSION

The VGPs derived from Jurassic igneous rocks of Antarctica are listed in Table 2 and plotted in Fig. 10. Most of them fall below latitude 60°S and are in good agreement with the Gondwana APW curve. The VGPs located closer to the pole have been regarded as anomalous. No experimental evidence has been found of either susbstantial secondary components or remagnetization, and the anomalous

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Table 2. VGP positions from Ferrar Supergroup rocks, Antarctica.

	Reference	Locality	VGP Lat. S	VGP Long. E
1	-- Turnbull, *1959*	Ferrar Glacier	58	218
2	-- Blundell and Stephenson, *1959*	Theron Mountains	54	224
3	-- Bull et al., *1962*	Wright and Victoria Valleys	42	220
4	-- Briden and Oliver, *1963*	Beardmore Glacier	59	211
5	-- Beck, *1972*	Dufek Massif and Forrestal Range	56	192
6	-- Lovlie, *1979*	Vestfjella, Kraul Mountains	42	226
7	-- Burmester and Sheriff, *1980*	Dufek Massif and Forrestal Range	60	223
8	-- Cherry and Noltimier, *1981*	David Glacier	55	222
9	-- Hoffman et al., *1984*	Queen Alexandra Range	50	226
10	-- Delisle and Fromm, *1984*	Litell Rocks	78	220
		Morozumi and Alamein Range	71	226
		DeGoes Ridge and Johnstone Glacier	75	122
11	-- Funaki, *1984*	Wright Valley	45	208
		Allan Hills	47	227
		Mt. Fleming	69	221
12	-- McIntosh et al., *1986*	Mesa Range	64	210
		Section Peak	49	195
		" "	54	210
13	-- Delisle and Fromm, *1989*	Roberts Butte	77	219
		Sequence Hills	58	220
14	-- This paper	Northern Prince Albert Mountains	48	226
		" " "	79	181

Number in the first column refers to Fig. 10; the position of VGP no. 5 (Beck 1972) results from an inappropriate tilt correction (see Burmester & Sheriff 1980).

positions have been interpreted according to various hypotheses.

Delisle & Fromm (1984, 1989) maintain that a Cretaceous thermal event reset remanence directions in the Rennick Graben region on the strength of three considerations: (a) mineralization owing to hydrothermal activity at the Litell Rock site; (b) agreement of the more southerly VGPs with the Cretaceous segment of the reassembled Gondwana APW path; (c) the Cretaceous remagnetization event suggested by Schmierer & Burmester (1986) as an explanation of the VGP position derived from volcanic and sedimentary rocks of the Cambro–Ordovician Bowers Supergroup, in the Bowers Terrane of Northern Victoria Land. McIntosh *et al.* (1986) reject several hypotheses and conclude that incomplete averaging of palaeosecular variation is the only acceptable, even if not attractive,

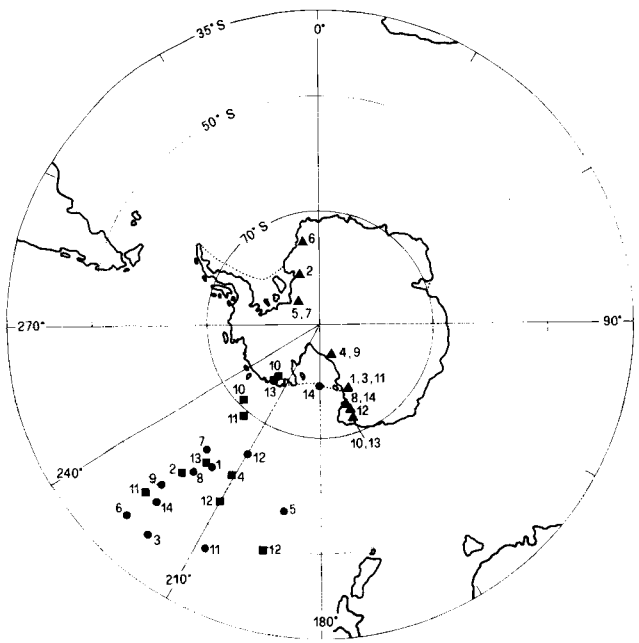


Figure 10. VGP positions of the Jurassic Ferrar Supergroup rocks. Symbols: circles = mean VGP; squares = VGP from single site or few samples; triangles = sampling locality; figures = reference as in Table 2.

explanation for the latitude dispersion (52°S to 72°S) of the site-mean VGPs from the Mesa Range.

Two significantly different VGP groups were found in this study of the northern Prince Albert Mountains. Taking as normal the VGPs surrounding the palaeopole (48°S, 226°E), the location of those falling at higher latitude (79°S, 181°E) requires an explanation. The hypothesis of a Cretaceous thermal event has to be examined first, because it has been advanced to explain both the palaeomagnetic results obtained by Schmierer & Burmester (1986) from the Bowers Terrane, and the wide dispersion of K-Ar ages found in the Kirkpatrick basalts from Mesa Range (Faure & Mensing 1993). Both studies suggest that the maximum temperature of the event was moderate, and Schmierer & Burmester give an estimate of $\leqslant 200\,^{\circ}C$.

The Ferrar dolerite samples from our anomalous sites were heated at 250 °C during the PTVRM experiment and the rate of remanence acquisition was too low to account for a complete remagnetization over a period of the order of magnitude of 1 Myr. An eventual reheating would therefore have only resulted in partial remagnetization, and the NRM should be made up of two components, corresponding to the blocked and unblocked part respectively of the Jurassic primary magnetization. Thermal demagnetization tests, however, disclosed only one component, stable throughout the blocking temperature spectrum up to the Curie point (Fig. 4a). No post-Jurassic thermal event can therefore be inferred from these data. The only reheating that might be supposed would have occurred soon after the time of sill intrusion, as, for example, the hydrothermal activity which affected the Ferrar Supergroup rocks studied by Hoffman *et al.* (1984) in the Queen Alexandra Range.

Lastly, the notion of a general thermal event extending from the Bowers Mountains and the Mesa Range to the Prince Albert Mountains is, in our opinion, hardly reconcilable with the field occurrence of the sites giving normal and anomalous VGPs. These sites are only a few kilometres apart (Fig. 1), whereas they are hundreds of kilometres far from the Bowers Mountains and the Mesa Range.

Following a suggestion by Delisle & Fromm (1984), a much younger, Cretaceous age of intrusion might be assumed for the sills with anomalous VGP. As reliable datings of Ferrar dolerite give ages between 180 and 165 Ma, this hypothesis is hardly tenable, because both chronological data from other magmatic provinces and model predictions (Campbell & Griffiths 1990) concur to suggest sudden onset and a few million years' duration for such continental magmatism.

Incomplete averaging of palaeosecular variation may explain the mean VGP from the Mesa Range (McIntosh *et al.* 1986) as the VGPs of the individual flows are randomly dispersed. In our case, however, palaeosecular variation explains only the scattering within each VGP group, and cannot be reconciled with the systematically higher latitude of the anomalous VGPs.

In the northern Prince Albert Mountains, the sites characterized by poleward displaced VGPs are confined to separate subregions (Fig. 1 and Table 1): the southern segment of Skinner Ridge and the nunataks to the south of the David Glacier, with the exception of Mount Joyce. Block tilting related to the uplift of the Transantarctic

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Mountains would therefore seem an attractive explanation. Unfortunately, no field evidence as yet supports the tilting of ≈25° needed to embrace all the VGPs.

Lastly, we may point out that no reverse or transitional polarities have as yet been reported from Ferrar dolerites from Victoria Land, except in some samples from our site Thern A. Magnetostratigraphical studies, on the other hand, have shown that many reversals occurred in the Middle Jurassic, the mean duration of the polarity intervals being ≈0.26 Myr (Steiner, Ogg & Sandoval 1987). Sill intrusion would have occurred in a very short time span and individual sills may well have recorded a part of the VGP path during a reversal. Fast and slow directional changes appear to alternate within a reversal (Laj, Guitton & Kissel 1987; Tric *et al.* 1991b), and in some instances VGPs are confined to small-radius loops (Herrero-Bervera *et al.* 1989). In the northern Prince Albert Mountains, fast geomagnetic variations may be inferred from the wide-latitude changes of the Thern A VGPs, and slow variations from the anomalous VGPs clustered around the palaeopole (79°S, 181°E). This palaeopole is shown as *P* in Fig. 11, together with the Jurassic palaeopole *J* from the pre-1984 literature data, and all the single-sample VGPs from Thern A, which plot approximately along the great circle PAM, drawn through *P* and *J*. As mentioned in the previous section, evidence has been found that the VGP path during the transitional phases of a reversal is often confined to a narrow longitudinal band. We may therefore assume PAM as a Jurassic palaeomeridian and all our anomalous VGPs as transitional.

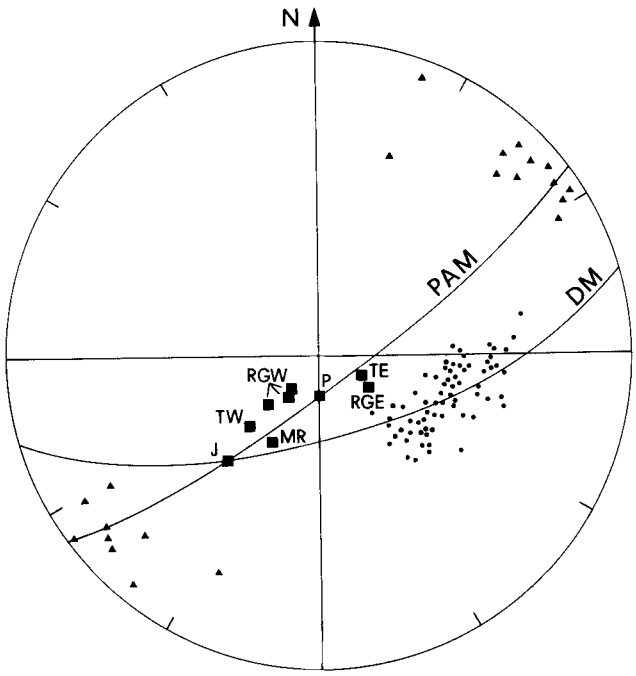


Figure 11. Equal-area projection of the Antarctica Jurassic VGPs deviating from the pre-1984 literature data palaeopole (see text for full discussion and references). Symbols: *J* = pre-1984 palaeopole; *P* = palaeopole from anomalous VGPs from Prince Albert Mts; triangles = single-sample VGPs from Thern A sill; dots = transitional VGPs from the Dufek Massif; PAM, DM = Prince Albert Mts and Dufek Massif palaeomeridian; MR = Mesa Range VGP; RGW, RGE = Rennick Graben VGP from westerly and easterly directions; TW, TE = Tasmanian dolerite VGP from westerly and easterly directions (rotated to Antarctica).

It is interesting to note that other anomalous Jurassic VGPs from Northern Victoria Land also plot near the great circle PAM (Fig. 11): (a) the mean VGP from the Kirkpatrick basalt in the Mesa Range (McIntosh *et al.* 1986), shown as MR. (b) the VGPs from the Rennick Graben (Delisle & Fromm 1984, 1989), shown as RGW and RGE. The RGE pole results from two Ferrar Dolerite sills characterized by ChRM declination directed eastward as opposed to westward in the other sites (RGW). The same duality was found in the Middle Jurassic Tasmanian dolerites (Schmidt & McDougall 1977), and the directions are similar to those in the Rennick Graben after reassembling Antarctica and Australia (Fig. 10 in Delisle & Fromm 1984). The VGPs derived from the western and eastern directions of Tasmanian dolerites are shown as TW and TE in Fig. 11, after having been rotated to Antarctica using the Euler pole 3.6°S, 40°E, rotation −31° (Smith & Hallam 1970). TW is not far from the Antarctica palaeopole *J*, and TE agrees well with RGE and plots near the great circle PAM.

A Jurassic transitional palaeomeridian is already known from Antarctica (Beck & Burmester 1979; Burmester & Sheriff 1980; Burmester & Beck 1981). Normal, reverse and transitional magnetizations have been found in the gabbroic rocks from the Dufek Massif (age 172 ± 4 Ma). The transitional VGPs from the Dufek Massif are shown in Fig. 11, together with the corresponding palaeomeridian DM, which is 30° away from the palaeomeridian PAM from the northern Prince Albert Mountains. The intrusion of the Dufek Massif and the Ferrar dolerite sills occurred during a time span of a few million years, and the transitional configuration of the geomagnetic field is supposed to persist over such time span (Clement 1991; Tric *et al.* 1991a; Constable 1992).

CONCLUSIONS

This palaeomagnetic study of the Ferrar dolerite in the northern Prince Albert Mountains has shown that all sites are characterized by single-component remanence, whose direction is stable up to the Curie temperature. No evidence of remagnetization was found, and PTVRM tests suggested that the ChRM acquired at the higher blocking temperature would not have been unblocked by subsequent heating at ≤200 °C, as in the Cretaceous thermal event proposed for rocks from the Bowers Terrane of Northern Victoria Land. The VGP position at 15 sites was consistent with the East Antarctica Jurassic palaeopole from the literature data, displaced toward higher latitudes at seven sites. One sill was characterized by large variations of the ChRM inclination, and the single-sample VGPs were located astride the equator.

These results raise two main questions. The first is to explain the position of the VGPs, which we interpret as the result of a slightly different age of intrusion and varying geomagnetic field. As the anomalous VGPs are situated along a great circle through the Jurassic palaeopole, we advance the hypothesis that they are transitional and the great circle corresponds to a Jurassic reversal path along a palaeomeridian. This hypothesis may also explain the position of the other anomalous VGPs from northern Victoria Land. The longitudinal band defined by the northern Prince Albert Mountains and the Dufek Massif

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palaeomeridians is comparable to the band found for Pliocene–Quaternary reversal paths.

The second question is how to estimate the best position of the East Antarctica Jurassic palaeopole. The position derived from pre-1984 literature data (53°S, 222°E) might be challenged on the grounds of the latitude dispersion of the Ferrar Supergroup VGPs from Victoria Land. Calculation of a new palaeopole giving unit weight to the VGPs listed in Table 2 could result in a biased position, because of the greatly varying number of sites examined in each study. We choose to give unit weight to individual-site VGPs, and collected from the literature as many VGPs as possible, calculating their position when the palaeomagnetic direction was only reported. Leaving aside the transitional VGPs from the Dufek Massif (Burmester & Sheriff 1980) and the easterly VGPs from the Rennick Graben (Delisle & Fromm 1984), a total of 103 VGPs are available, and the calculated Jurassic palaeopole is 57°S, 221°E ($K = 24$, $A_{95} = 2.9°$).

The VGPs displaced towards the geographic pole have been regarded as anomalous (Delisle & Fromm 1984, 1989; McIntosh *et al.* 1986; this paper), irrespective of the hypothesis put forward to explain the anomaly (remagnetization, palaeosecular variation, transitional geomagnetic field). It is therefore advisable to cut the anomalous VGPs off and calculate a corrected palaeopole. The value of 65°S is the lowest latitude threshold that may be used to cut-off (Fig. 10). The corresponding palaeopole, calculated from 77 VGPs, is 51°S, 223°E ($K = 36$, $A_{95} = 2.7°$). The Jurassic palaeopole of East Antarctica therefore lies between two extreme points: (57°S, 221°E), when all VGPs are given the same significance and (51°S, 223°E), when the VGPs located south of 65°S are considered as anomalous.

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